

A Two-Step Ensemble-Based Genetic Algorithm for Land Cover Classification

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Abstract—Accurate land use and land cover (LULC) maps are effective tools to help achieve sound urban planning and precision agriculture. As an intelligent optimization technology, genetic algorithm (GA) has been successfully applied to various image classification tasks in recent years. However, simple GA faces challenges, such as complex calculation, poor noise immunity, and slow convergence. This research proposes a two-step ensemble protocol for LULC classification using a grayscale-spatial-based GA model. The first ensemble framework uses fuzzy c-means to classify pixels into those that are difficult to cluster and those that are easy to cluster, which aids in reducing the search space for evolutionary computation. The second ensemble framework uses neighborhood windows as heuristic information to adaptively modify the objective function and mutation probability of the GA, which brings valuable benefits to the discrimination and decision of GA. In this study, three research areas in Dangyang, China, are utilized to validate the effectiveness of the proposed method. The experiments show that the proposed method can effectively maintain the image details, restrain noise, and achieve rapid algorithm convergence. Compared with the reference methods, the best overall accuracy obtained by the proposed algorithm is 88.72%.

Index Terms—Genetic algorithm (GA), land use and land cover (LULC), neighborhood window, remote sensing image classification, two-step ensemble.

I. INTRODUCTION

LAND use and land cover (LULC) plays an essential role in various aspects, including ecosystem, urban planning,

and social economy, of profound importance in the Earth system [1], [2], [3], [4]. Field-based studies, relying on manual interpretation, are time-consuming and labor-intensive manners that cannot classify the land-use status in time [5], [6], [7]. On the contrary, recent progress in satellite technology provides excellent potential for accurately distinguishing between diverse land features characteristics and rapidly creating LULC maps of the vast area [8], [9], [10]. The remote sensing image classification is roughly divided into supervised and unsupervised approaches [11], [12], [13], [14]. The superior performance of supervised classifiers substantially depends on the high-cost labeled data, which is difficult to obtain [15]. Unsupervised classifiers obtain a data distribution by dividing the image into pixel groups with similar characteristics, such as K-means [16], ISODATA [17], and fuzzy c-means (FCM) [18], which have been successfully utilized in the LULC classification domain. For example, Dang et al. [19] proposed the multiple kernel collaborative FCMs clustering with weighted super-pixel granulation technique, which can reduce the considered objects of land-cover classification from a large amount of pixels to only a few hundred super-pixels amount of the considered objects. Paradis relaxed the assumption of homogeneous variance in the standard k-means algorithm so that presented a novel land cover classification method called probabilistic k-means (P-Kmeans) [20], this method performs well on noise multi-spectral data and has a substantial improvement in the overall accuracy (OA) compared to the simple k-means. Unfortunately, predictions of these classifiers only consider spectral features and ignore their spatial correlation, leading to poor robustness.

The joint spatial-spectral feature representation has been found to improve algorithm noise immunity and reduce classification errors in many existing classification models [21], especially fuzzy clustering algorithms [22], [23], [24]. FLICM [25] uses fuzzy space and grayscale similarity measurements to enhance image detail retention while ensuring noise insensitivity. Zhang et al. [26] enhanced the conventional FCMs algorithm using adaptive weighting factors for neighboring pixel effects based on a novel pixel spatial attraction model. Lei et al. [27] proposed simpler and significantly faster FRFCM by introducing morphological reconstruction operation and local membership filtering. Wang et al. [28] presented FCM_SICM with adaptive spatial intensity constraint and membership linking, reducing the time complexity of objective function from $O(n^4)$. However, improved FCM algorithms still present challenges in terms of noise interference reduction.

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It has been shown that the land classification problem can be decomposed into multiple binary classification tasks [29]. Finding an optimal classification is, hence, considered a combinatorial optimization problem. During the last decade, there has been a remarkable breakthrough in the research of land use classification based on evolutionary computation [30], [31]. For example, Yang et al. presented a particle swarm optimization (PSO)-based technique and designed a criterion function to identify the land cover [32]. Sarkar et al. [33] used the maximum R nyi entropy (MRE) to develop an approach for unsupervised classification of land cover based on differential evolution (DE), which perform well on two widely used hyperspectral image data. Genetic algorithms (GAs), effective global optimization algorithms, imitate biological genetic mechanisms in the process of natural selection for determining a globally optimal solution [34], [35], [36], [37]. The standard GA requires a binary encoding operation for representing a candidate solution in the form of a string [38], [39], [40]. In each iteration, the crossover operator randomly selects two candidate solutions (i.e., parents) and exchanges part of their information to create new solutions (i.e., offspring), which enhances the exploration capability of the population [41]. Like the gene mutation in nature, the mutation operator is applied to the GA to change part of the information in the offspring solutions and enhance the local searchability [42]. The selection operator selects the candidate solutions based on the fitness of each member of the population—reserving better progenies and eliminating bad progenies. This mechanism, in turn, enhances the balance between exploitation and exploration [43]. By the abovementioned genetic operators, these individuals are combined to produce a new generation of candidate solution groups, and the process is repeated until some convergence index is satisfied. GA acquire and accumulate knowledge about the search direction during the search process and, thus, automatically control the search direction to generate the optimal solution [44], which has been successfully applied to remote sensing image classification and have achieved good results. Many improved GAs have emerged in recent years [45]. However, there are still unresolved issues.

- 1) *Extensive calculations.* The convergence of GA deteriorates when the dataset becomes larger.
- 2) The poor antinoise performance simple GA focusing on spectral features is usually sensitive to noises in the image.
- 3) *Premature convergence.* GA has limited search capability for the new space and is prone to fall into local optimum.

As the remotely sensed data is characterized by heterogeneity, we focus on ensemble classifiers, which emphasize complementarity among the various classification features [46], [47], [48]. The critical idea of ensemble strategy is to blend and combine predictions from different classification rules [49], [50]. In the study of [51], a highly accurate ensemble classification model is proposed to provide multiple independent classification maps by different feature descriptors. The study of [52] indicates that the ensemble framework consisting of a two-step voting mechanism can suppress the noise of the original image data and refine sea ice classification to some extent.

In this article, a two-step ensemble strategy-based improved GA model (TE-nGA) is proposed for land cover classification of

multispectral images. For the deficiencies of GA, our solution is to use ensemble strategy and spatial windows on top of a classical GA to improve antinoise performance, classification accuracy, and convergence speed. Major contributions of this article are as follows.

- 1) The FCM algorithm is used to classify pixels into those that are difficult to cluster and those that are easy to cluster, which narrows down the computational complexity.
- 2) A grayscale-spatial-based GA is proposed by considering the spatial neighborhood window in heuristic parameter-making to prevent converging prematurely and noise-sensitive.
- 3) A novel two-step ensemble framework with a multiband classification mechanism is used to reduce the instability of the solution and classification errors.

II. PROPOSED CLASSIFICATION METHOD

The flowchart of the proposed algorithm is shown in Fig. 1. And the algorithm mainly consists of the following three steps: image denoising, finding the pixels difficult to cluster using voting strategy, and reclassification by improved GA based on ensemble model and spectral–spatial fusion. Note that, the second and third steps are the critical parts of the proposed algorithm. The proposed method is sketched in Algorithm 1.

A. Image Denoising

To avoid image artifacts from image sensor noise that affects the quality of the scene, the algorithm consists in first applying block-matching and 3-D filtering, an existing popular and effective denoising algorithm [53]. The characteristics of this filtering are chosen based on this general information, it is a tradeoff between reducing the remaining noise in the rectified image and preserving the changes in intensity among neighboring pixels belonging to different clusters. The algorithm has been proven to outperform other filtering algorithms in improving image quality.

B. First Ensemble Stage—Finding Pixels Difficult to Cluster

The block diagram on the left of Fig. 1 shows a multispectral image with D wavelengths, from which can extract D specific images with size of $H \times W$, denoted as $I_1, I_2, \dots, I_D$. The first step, partitions pixels between those that are difficult to cluster and those that are easy to cluster in that at least for one spectral image, their membership is very likely. Therefore, the FCMs clustering is being applied separately on each spectral image to classify pixels into N categories, where N is set in advance by a visual interpretation. It produces for each pixel, each wavelength, and each cluster a membership

$$u_d^n(i, j) = \text{FCM}[(I_d(i, j))], 1 \leq i \leq H, 1 \leq j \leq W \quad (1)$$

where $I_d(i, j)$ is the intensity of the pixel (i, j) at the d th wavelength, $u_d^n(i, j)$ is the membership to the n th cluster of the pixel (i, j) at the d th wavelength, $1 \leq d \leq D, 1 \leq n \leq N$. For the d th wavelength, each pixel is marked with a category

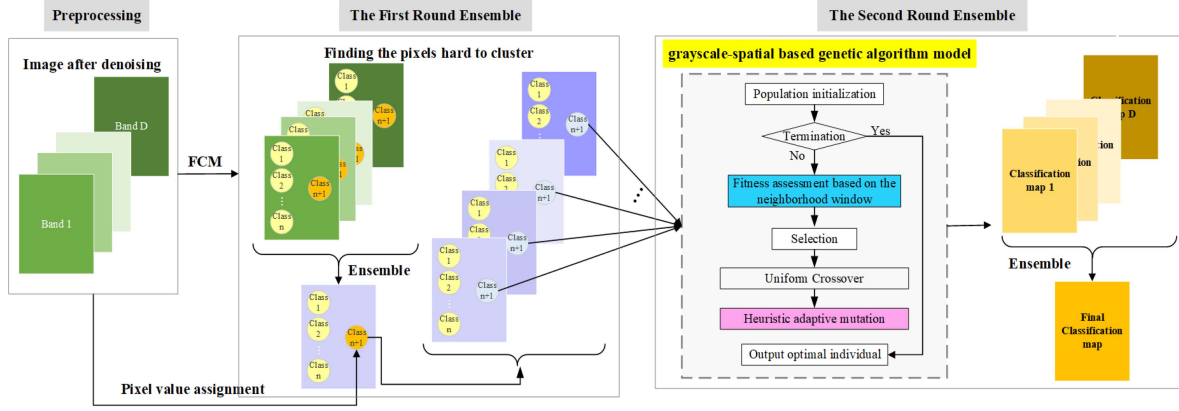


Fig. 1. LULC classification flowchart based on the proposed method.

label $c_d(i, j)$, as

$$\begin{cases} \text{if } \exists n, \max \{u_d^n(i, j)\} \geq \tau, & c_d(i, j) = \arg \max_{n \leq N} \{u_d^n(i, j)\} \\ \text{if not,} & c_d(i, j) = N + 1 \end{cases} \quad (2)$$

based on the membership, pixels are classified as easily clustered $\mathcal{K}_e$ and hard to cluster $\mathcal{K}_d$. The threshold τ used is a parameter to ensure that the pixel $(i, j) \in \mathcal{K}_e$ considered easy to cluster has a high degree of membership with the relevant cluster center. The $(N + 1)$ th cluster contains the pixels considered difficult to cluster and denoted $\mathcal{K}_d$.

Voting-based approaches can resolve conflicts between different decisions generated by multiple classification rules. Accordingly, the preclassification results of each wavelength obtained by FCM are integrated into the ensemble preclassification map C , based on the majority voting principle. The pixel label in C is remapped to the corresponding position of the filtered multispectral image, and the final reclassification maps are obtained, denoted as $I'_1, I'_2, \dots, I'_D$, as the frame shown in the middle of Fig. 1. After applying the FCM algorithm to preclassify the pixels as easy and difficult to cluster, the issue of analyzing the entire original image is, thus, converted into the classification of the difficult pixels.

C. Second Ensemble Stage—Improved GA Taking Into Account Neighboring Pixels

On the right of the flowchart, a frame entitled “The Second Round Ensemble” figures two tasks: GA classification based on spectral-space joint factor and decision ensemble for multiband classification. Since the LULC classification problem is decomposed into multiple binary classification tasks, we should address series subproblems for each wavelength: whether the pixel $(i, j) \in \mathcal{K}_d$ belongs to the x th category? $1 \leq x \leq N$. Finally, the decision results of all N binary classification maps are combined to obtain a classification map containing N land cover types. For each binary classification problem, the improved spectral-space-based GA is applied to pixels marked as the $(N + 1)$ th class to optimize further the ambiguous category labels, illustrated in the dashed frame of Fig. 1. The interpretation of this improved spectral-space-based GA is as follows.

1) *Two-Dimensional Binary Chromosome Encoding*: For image I'_d , N binary maps can be extracted from N classification tasks. Two-dimensional binary encoding is utilized to maintain the spatial structural properties of evolved individuals (one individual represents one potential decision). The encoding matrix size is the same as the wavelength image, that is $H \times W$. To estimate whether pixel $(i, j) \in \mathcal{K}_d$ belong to the x th category, $1 \leq x \leq N$, for each wavelength, we encode pixels $(i, j) \in \mathcal{K}_e$ in class x clustered by the FCM as 1 and noncategory x as 0, where no evolution optimization operation is performed further. Those points $(i, j) \in \mathcal{K}_d$ are randomly encoded as 0 or 1. Evolutionary optimization implemented in $\mathcal{K}_d$ will explore all possible solutions.

2) *Fitness Function Based on Neighborhood Window*: Fitness function is the most critical indicator to evaluate the quality of individuals in the population, which is related to the selection of individuals and the execution of genetic operators. Adjacent pixels are highly correlated in terms of grayscale and texture. Consequently, the complementary information of spectra and space is adopted to develop a composite fitness function, which provides a novel idea for improving algorithm noise immunity.

For the n th clustering task at the d th wavelength, suppose the number of individuals in the population is P , $CI_{p,n,d}$ is the p th individuals, $1 \leq p \leq P$, $1 \leq n \leq N$, $1 \leq d \leq D$. The fitness function of $CI_{p,n,d}$ is $\text{Fitness} = \frac{1}{\text{OF}}$, OF is calculated by formula (3)

$$\text{OF} = \frac{1}{\sum_{r=0}^1 N_r} \sum_{r=0}^1 N_r \sum_{(i,j) \in R_r} \left[W_r(i, j) + (I'_d(i, j) - v_r)^2 \right] \quad (3)$$

$$W_r(i, j) = \frac{1}{L} \sum_{t \in N(i, j, s)} \frac{(N_t(i, j, s) - v_r) \times u_{tr}(i, j, s)}{l_t(i, j, s) + 1} \quad (4)$$

$$u_{tr}(i, j, s) = \frac{(N_t(i, j, s) - v_r)^2}{\sum_{r=0}^1 (N_t(i, j, s) - v_r)^2} \quad (5)$$

$$L = \sum_{t \in N(i, j, s)} \frac{1}{l_t(i, j, s) + 1} \quad (6)$$

where $r \in \{0, 1\}$, R_r indicates the pixels encoded as 0 or 1, and their quantity statistics are represented as N_r . The r th cluster center v_r is calculated by the pixel mean of all points belonging to R_r . $I'_d(i, j)$ is the intensity of the pixel (i, j) at the d th wavelength after applying the FCM clustering algorithm.

In the neighborhood function $W_r(i, j)$, neighborhood windows are assumed to be square, and their side length s to be odd integers. $N(i, j, s)$ represents the $s \times s$ neighborhood window at indexed location (i, j) . $N_t(i, j, s)$ stands for the intensity of the t th pixel in the neighborhood $N(i, j, s)$. The membership degree of the t th pixel in $N(i, j, s)$ belonging to the r th cluster is denoted as $u_{tr}(i, j, s)$. $l_t(i, j, s)$ represents the spatial distance between the t th neighborhood pixel and the center point (i, j) . L is the spatial distance weight between the center point and its neighborhood pixels.

3) *Mutation Operator Using Neighborhood Information:* New generations are being iteratively generated via individual selection, uniform crossover, and genetic mutation, as illustrated in the dashed frame of Fig. 1. Most GAs utilize a global mutation probability to implement the label modification for randomly selected pixels with the same probability, which invisibly slows down the convergence rate of GA. As the critical technical operator, an improved heuristic mutation probability, in combination with neighborhood information, is created to automatically calculate the appropriate mutation probability for each gene. The following formulas define this novel mutation probability:

$$u_{ht}(i, j, s) = \frac{1}{\sum_{r=0}^1 \frac{(N_t(i, j, s) - v_h)^2}{(N_t(i, j, s) - v_r)^2}} \quad (7)$$

$$P_m(i, j) = P_0 \times S^\sigma \quad (8)$$

$$S = \text{CI}_{p,n,d}(i, j) \times \sum_{t \in N(i, j, s)} \frac{1 - C_t}{l_t(i, j, s)} \quad (9)$$

for the center pixel (i, j) , $u_{ht}(i, j, s)$ denotes the membership degree of neighborhood point $N_t(i, j, s)$ belonging to clustering center v_h , $h \in \{0, 1\}$.

Consequently, the adaptive mutation probability $P_m(i, j)$ of the individual $\text{CI}_{p,n,d}$ is expressed as formulas 8, where P_0 is the basic mutation probability, also the global mutation probability in classical GA; σ , the exponent of S , stands for the influence weight of neighborhood information on the mutation probability of the central pixel. $\text{CI}_{p,n,d}(i, j)$ is the binary encoding of the p th individual, for the n th clustering task at the d th wavelength. $l_t(i, j, s)$ indicates the spatial distance between the center pixel (i, j) and the t th pixel in $N(i, j, s)$. The label of $N_t(i, j, s)$ is denoted as C_t , if u_{1t} is greater than u_{0t} , then $C_t = 1$; otherwise, $C_t = 0$. Considering the different neighborhood windows, when $s = 3$, it can be calculated that $S \in [0, 4 + 2\sqrt{2}]$; when $s = 5$, $S \in [0, 6 + 3\sqrt{2} + 1.6\sqrt{5}]$.

Through the abovementioned mechanism, points $(i, j) \in \mathcal{K}_d$ can adaptively acquire appropriate mutation probabilities based on the category of neighboring pixels. When the estimated label $\text{CI}_{p,n,d}(i, j)$ is in agreement with most categories of surrounding pixels, a small mutation probability is required to maintain the original regional homogeneity. Otherwise, the mutation probability increase to make $\text{CI}_{p,n,d}(i, j)$ tend to most categories

Algorithm 1: The Procedure of the Proposed Two-Step Ensemble Model.

- 1: **Input:** multi-spectral image with a size denoted as $H \times W \times D$, membership threshold τ , number of individuals P , the size of neighborhood window s , basic mutation probability P_0 , crossover probability P_c , weight of the neighborhood σ , the counter of generation g , the number of maximum generation G .
 - 2: **The first ensemble phase:**
 - 3: **for** $l=1:D$ **do**
 - 4: FCM and τ are applied separately on each spectral image to classify pixels as $\mathcal{K}_e$ and $\mathcal{K}_d$.
 - 5: **end for**
 - 6: D pre-classification maps are integrated into an ensemble pre-classification map.
 - 7: **The second ensemble phase:**
 - 8: **for** $i=1:D$ **do**
 - 9: **for** $j=1:N$ **do**
 - 10: **loop**
 - 11: Initialize P individuals, each is encoded as a two-dimension binary matrix, $g = 1$.
 - 12: Select individuals based on fitness with the elite retention strategy.
 - 13: Uniform crossover (using P_c) is applied to the selected individuals to swap genes for pixels in $\mathcal{K}_d$.
 - 14: The heuristic mutation operator (using P_0 and weight coefficient σ) is developed to modify genes for pixels in $\mathcal{K}_d$.
 - 15: $g = g + 1$, if $g > G$, break the infinite loop. Otherwise, go to Step 11.
 - 16: **end loop**
 - 17: Output the individual with the highest fitness.
 - 18: **end for**
 - 19: Combine the N individuals into an N -classified map.
 - 20: **end for**
 - 21: Decision integration of the D classification maps and output of the final classification maps.
-

in the neighborhood $N(i, j, s)$. By the abovementioned genetic operators, new offspring are being iteratively generated. The individual with the highest fitness is output when the iteration condition is not satisfied, as shown in steps 10–16 of Algorithm 1.

Note an unusual classification rule of this proposed method in terms of LULC classification. The proposed spectral-space-based GA performs only binary classification on each wavelength image. Thus, when all N land use types have been discriminated and obtain N binary images, the classification results are fused into a complete classification map. In simple terms, pixel (i, j) belongs to the category that marks it as 1 in the binary classification map. When two or more binary maps mark the pixel as 1, the pixel is classified as the one with the larger fitness value, as shown in Fig. 2 and step 19 of Algorithm 1.

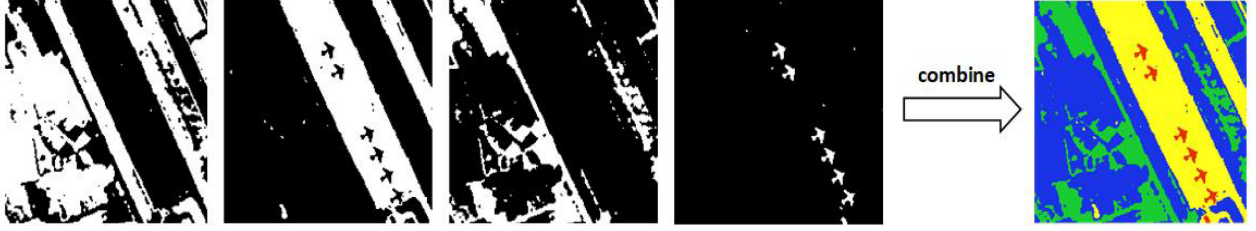


Fig. 2. Combination of four binary classification result maps.

Finally, the second ensemble model is applied to the D classification maps, aiming to merge classification maps from all wavelengths into the final LULC map. The ensemble rules are as follows: suppose that there are n_k classification maps classifying the pixel (i, j) as the k th class, and the final category $CI(i, j) = k$, if $n_k > n_x$, where $x \in \{1, 2, \dots, N\}$, $x \neq k$, $\sum_{i=1}^N n_i = D$. Note that in the integration framework, we always selectively remove the classification results in the NIR band, because its classification results are often incredible.

III. EXPERIMENTAL RESULTS

A. Introduction to Data

Three datasets of remote sensing images acquired by the Beijing 2 satellite are used to verify the accuracy and effectiveness of the proposed algorithm. Beijing-2 is a civil commercial remote sensing satellite constellation (DMC3) consisting of three high-resolution satellites. It was launched on July 11, 2015 from India and delivered to a 651 km sun-synchronous. The Beijing-2 constellation system provides approximately 24 km wide, 0.8 m resolution panchromatic, and 3.2 m resolution blue, green, red, and near-infrared multispectral images in orbit, which can provide remote sensing satellite data and spatial information products with global coverage and excellent spatial and temporal resolution, capable of revisiting any location on the globe for one or two day. Experimental images were obtained in 2018. The test site is three different parts of Dangyang City, Hubei Province, China. In this article, only multispectral images are considered. We performed atmospheric correction on these image data through ENVI(5.3). The atmospheric model is subarctic winter and the aerosol model is rural. Moreover, the aerosol retrieval we selected is 2-band (K-T). The other parameters were set by default. All images are preprocessed by coregistration. Besides, we have been provided with an already labeled classification of all images in the four or five classes. This is based on an exhaustive visual inspection by observing the original image, and recorded on a georeferenced map as ground truth.

B. Parameter Configuration

In this experiment, the threshold τ of FCM is 0.8, the population size is set to 40, the crossover probability $P_c = 0.8$, the base mutation probability $P_0 = 0.001$, the exponent σ is 2, the neighborhood window size s is set at 3 pixels, aiming to maintain image details while also optimizing computer processing time

and convergence direction. Nine existing land cover classification approaches are needed to verify the effectiveness of the proposed method, including FCM, ENFCM [22], FGFCM [23], FLICM [25], ADFLICM [26], FRFCM [27], FCM_SICM [28], MRE-DE [33], and P-Kmeans [20]. The parameter settings are identical to the original literature, not reiterate them here. Furthermore, simple GA has the same parameter population size H and crossover probability P_c as the proposed algorithm in this article, the mutation probability $P_m = 0.01$, but without neighborhood information, the reader can understand that the neighborhood window size is 1. In particular, nGA retains the genetic operator using neighborhood information but deletes the ensemble operation existing in TE-nGA.

C. Classification Results

Figs. 3–5 show LULC classification results in color maps corresponding to the different models when tested on each dataset. Since the three remotely sensed images contain different types of land use, the color representation of the classification chart is not uniform. According to these classification results, for three figures, the color maps obtained by the proposed method are more clear and smooth.

Table I shows times (running time) and OA achieved by the 12 models and when tested on research areas to compare the classification performance with different methods. The best experimental results are shown in bold. One can conclude that the proposed method presented the best overall results, with the highest OA of 88.72% and relatively few wrong pixels. For Area 3, when using the neighborhood windows, the classification accuracy of LULC is increased by 28.35%; More importantly, TE-nGA can reduce the computational time to 9257.56 s, compared to the simple GA. It is because the heuristic mutation operator of the TE-nGA can adaptively control the search direction to accelerate the convergence of the algorithm, proving that the proposed method is more suitable for practical scenarios. In addition, nGA is equivalent to TE-nGA removing ensemble operation, the reader can understand that the TE-nGA runs only on the single-band (band1) image. For Area 3, when ensemble frameworks are introduced, the classification accuracy is further improved by 0.72%, demonstrating the ensemble strategy's effectiveness.

D. Parameter Analysis

One of the essentials of the proposed method is to regard the spatial neighborhood information as heuristic information

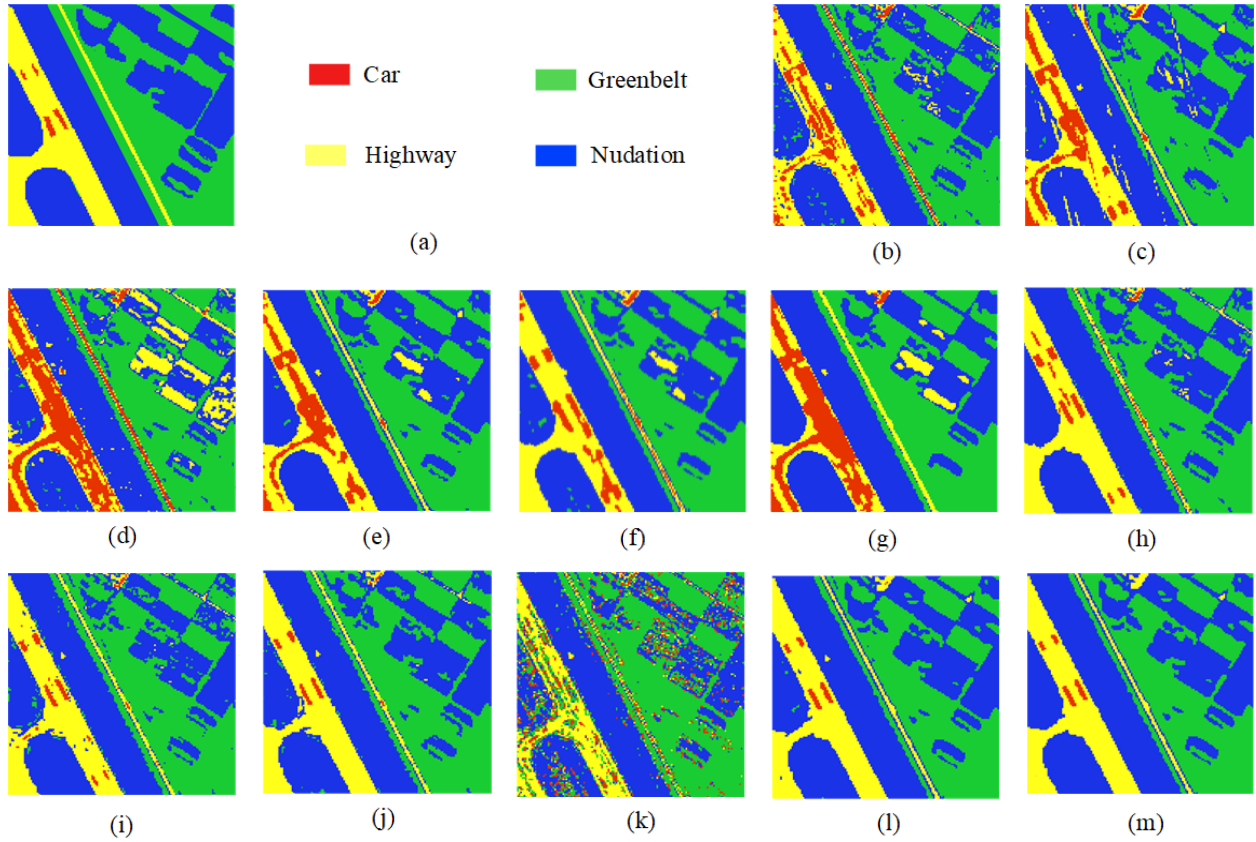


Fig. 3. Area 1 classification results. (a) Ground Truth. (b) FCM. (c) ENFCM. (d) FGFCM. (e) FLICM. (f) ADFLICM. (g) FRFCM. (h) FCM_SICM. (i) MRE-DE. (j) P-Kmeans. (k) GA. (l) nGA. (m) TE-nGA.

TABLE I
EVALUATION RESULTS OF EIGHT DIFFERENT CLASSIFICATION METHODS IN THREE STUDY AREAS

No.	Methods	Area1		Area2		Area3	
		<i>OA</i> (%)	<i>Time</i> (s)	<i>OA</i> (%)	<i>Time</i> (s)	<i>OA</i> (%)	<i>Time</i> (s)
1	FCM	83.27	0.71	76.02	0.73	68.21	1.10
2	ENFCM	81.83	0.08	77.81	0.11	68.98	0.12
3	FGFCM	72.63	0.16	76.84	0.24	59.27	0.28
4	FLICM	83.46	5.98	83.86	9.61	69.83	17.45
5	ADFLICM	85.54	1.23	78.03	3.24	77.92	3.89
6	FRFCM	79.92	0.19	86.90	0.21	81.83	0.26
7	FCM_SICM	86.30	3.21	78.43	8.36	69.79	11.19
8	MRE-DE	85.94	4258.28	77.02	5021.71	80.04	13088.64
9	P-Kmeans	87.81	3.13	78.68	8.27	82.11	9.29
10	GA	73.21	52711.02	63.95	68413.63	57.28	89768.13
11	nGA	88.49	1789.36	85.32	2653.19	84.91	9179.42
12	TE-nGA	88.72	1832.23	85.87	2685.46	85.63	9257.56

to adaptively modify the objective function and mutation probability, thus improving the noise immunity and classification accuracy, as well as convergence speed. However, it does not mean that the larger the window size, the better the classification effect. With respect to Table II, increased performance is observed when considering spatial neighborhood information, when increasing the size of the window up to an appropriate threshold, and then decreasing. All is for that samples farther away from the central pixel contribute little to the category

division of central pixel and even aggravate the misclassification. Due to objects at diverse geographical locations existing differences in brightness and texture, the optimal window size seems to be 5×5 for Area 1 and Area 2, yet 3×3 for Area 3. This result can be observed in bold in Table II, representing the best accuracy and fastest convergence time.

Furthermore, we have found that base mutation probability P_0 and the exponent σ are also vital factors affecting classification performance. In our study, different P_0 and σ are used to verify

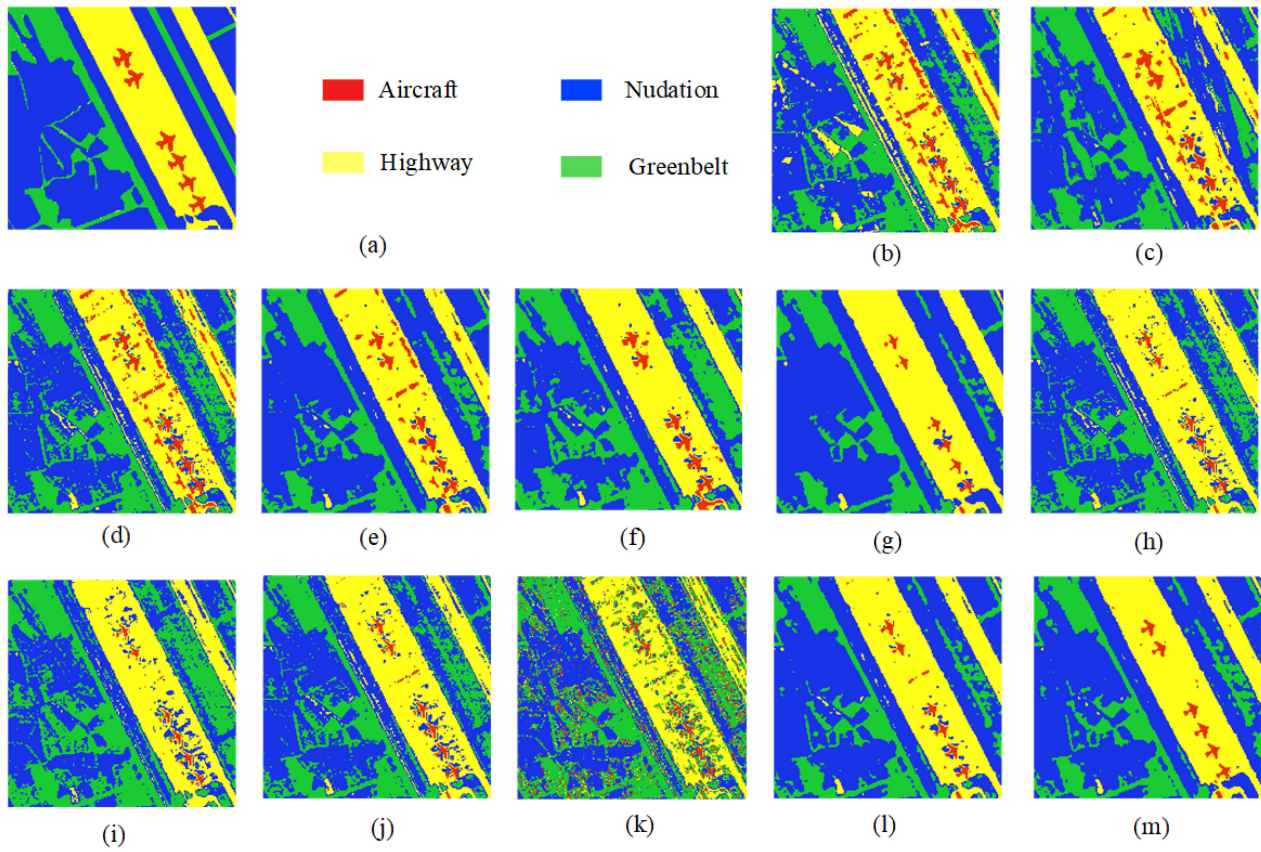


Fig. 4. Area 2 classification results. (a) Ground Truth. (b) FCM. (c) ENFCM. (d) FGFCM. (e) FLICM. (f) ADFLICM. (g) FRFCM. (h) FCM_SICM. (i) MRE-DE. (j) P-Kmeans. (k) GA. (l) nGA. (m) TE-nGA.

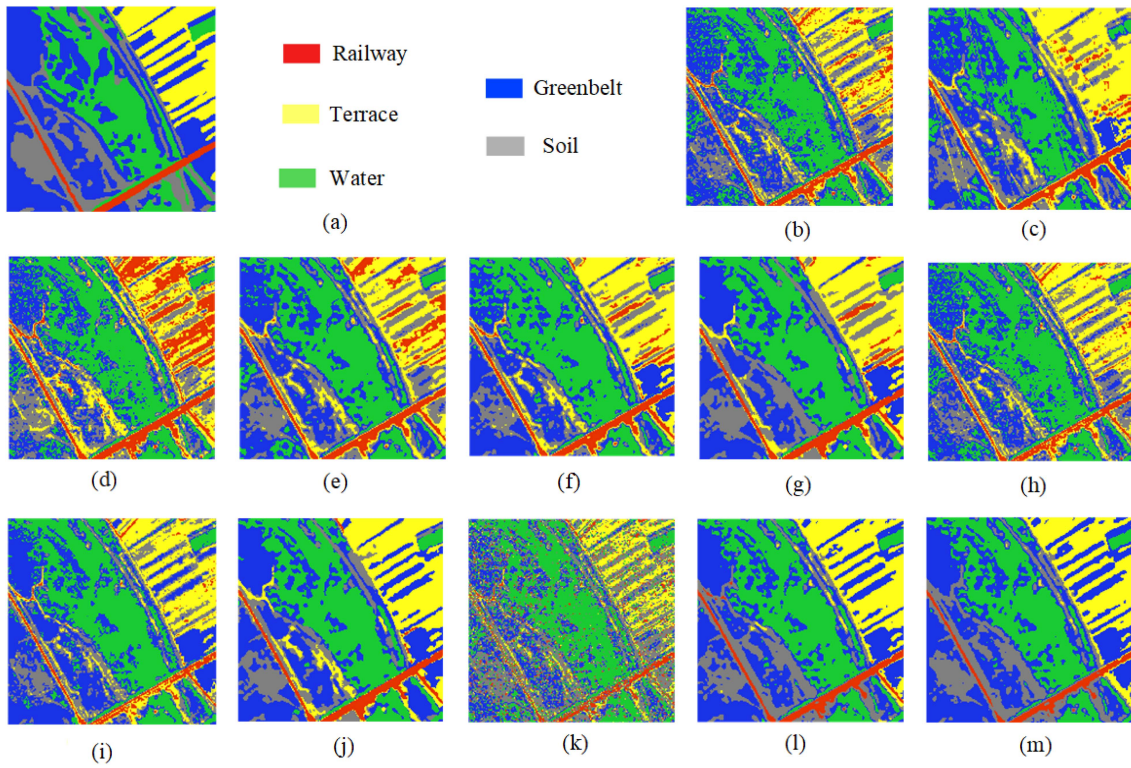


Fig. 5. Area 3 classification results. (a) Ground Truth. (b) FCM. (c) ENFCM. (d) FGFCM. (e) FLICM. (f) ADFLICM. (g) FRFCM. (h) FCM_SICM. (i) MRE-DE. (j) P-Kmeans. (k) GA. (l) nGA. (m) TE-nGA.

TABLE II
EVALUATION RESULTS OF THE PROPOSED TE-nGA USING DIFFERENT SIZE WINDOWS IN THREE STUDY AREAS

Study Area	3×3		5×5		7×7		9×9	
	OA(%)	Time(s)	OA(%)	Time(s)	OA(%)	Time(s)	OA(%)	Time(s)
Area1	88.72	1832.33	89.35	3668.91	84.32	5016.58	74.96	8113.98
Area2	85.87	2685.46	87.46	4219.37	79.92	5931.84	70.88	9831.84
Area3	85.63	9257.56	81.15	12257.56	75.39	20125.81	68.53	35970.32

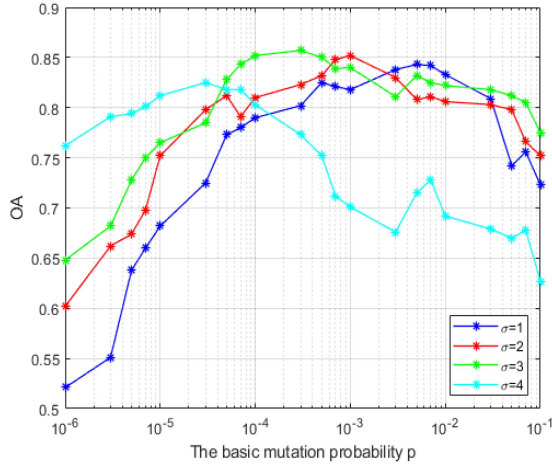


Fig. 6. Analysis of the base mutation probability P_0 and exponent σ in the heuristic mutation probability.

the interpretation and classification of LULC. Fig. 6 draws the variation curves of OA as P_0 and σ increase. It turned out that, for a fixed σ , as P_0 increases, the overall classification accuracy first increases and then decreases. This result is expected since a small P_0 will cause the search to be limited to the original evolutionary direction and, thus, fall into a local optimum; conversely, a large P_0 will break the good evolutionary pattern and deviate the results from the optimal solution. On the other hand, as σ increases, the effect of the neighborhood information is more significant, and the best basic probability value gradually changes from high to low. Suitable neighborhood weights are more conducive to obtaining the best classification results. In our research area, the best exponent is $\sigma \in \{2, 3\}$, the most apposite range of P_0 is $[10^{-5}, 10^{-2}]$.

IV. CONCLUSION

The main contribution of this study is proposing a superior two-step ensemble strategy and spectral-spatial-based GA for LULC classification of multispectral imagery. The first ensemble model performs FCM on multiband images, reducing the classification task of remotely sensed images to category determination of pixels difficult to cluster. The second ensemble processing comprises two tasks. First, the improved GA combining the spectral and spatial information was created to offer innovative solutions to puzzles, such as slow convergence and poor antinoise. Exactly, this improvement is benefited from the novel fitness functions and mutation probability, using an appropriate neighborhood window. Then, the ensemble model is adopted to integrate classification results derived from multiple

band images, which is used to reduce the risk of misclassification. The results prove that the proposed TE-nGA has preferable classification performance and faster convergence than simple GA. In the future, we expect to extend the application of the proposed technique to other fields, such as crop classification, target detection, and hyperspectral image classification.

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